

Event Detection and Video Summarization

Second Assignment — Computer Vision 2026

Group 9

TVSUM Videos 33–36

1 INTRODUCTION

This report documents our pipeline for the second assignment, in which we were assigned TVSUM videos 33–36: a graduation-day flash-mob speech (33), a street flash mob for the ICC World T20 (34), an in-train classical-orchestra advertisement (35), and a beekeeping interview (36). The four videos were chosen deliberately to stress-test our models across very different visual styles—choreographed performance, outdoor crowd events, a confined public-transport setting, and a single-speaker documentary.

Each video was annotated independently by both group members before either annotation was shared. We then built a six-step pipeline: measuring inter-annotator agreement, detecting events automatically with a Vision-Language Model, evaluating how well those detected events cover our annotations, localising each event in the video with two moment-retrieval methods, measuring localisation accuracy with IoU, and finally assembling a concatenated summary video per recording. All code lives under `pipeline/` and is invoked from the accompanying Jupyter notebook `Group9_Second_Assignment_CV2026.ipynb`.

2 ANNOTATIONS AND INTER-ANNOTATOR AGREEMENT

We followed the prescribed format—event description, timestamps formatted as MM:SS – MM:SS, and a subjectivity score—and produced between 8 and 15 events per video. Both annotators watched each recording in full before writing anything down, and only compared files once both were complete, so the agreement figures reflect genuine independent judgement rather than negotiated consensus.

Subjectivity choices

Deciding which score to assign each event was the most deliberate part of the annotation process. We used the full $-2 \dots +2$ scale and tried to apply it consistently across all four videos. A rating of $+2$ went to moments that are structurally indispensable—events a viewer could not skip without losing the thread of the video. The first musician revealing herself on the train in video 35 (00:39–00:45) is a good example: without that moment the flash-mob concept never becomes clear. Similarly, the beekeeper opening a hive for the first time in video 36 (00:59–01:06) is the visual anchor around which everything else in that video hangs. Ratings of $+1$ went to events that add meaningful context without being strictly necessary—the conductor stepping forward in video 35 fits here because his presence enriches the scene but the music is already self-explanatory by that point. We assigned zero to events whose relevance depends on whether the viewer already knows something about the subject: a baby watching the musicians (video 35) is charming, but it is not load-bearing for the narrative. Negative ratings marked events we included because they colour the summary yet are not required to

follow the main argument—the crowd laughing at an anecdote, or a close-up of a single honeycomb frame.

This subjectivity had a direct downstream effect. Video 33, the graduation speech, produced the widest disagreement: one of us marked ten events, the other nine, and the scores diverged most on the speech segments, where the emotional weight of the words is invisible to any purely visual pipeline. That mismatch foreshadows the moment-retrieval results in Section 4.

Agreement metrics

We quantified agreement along three complementary dimensions. The **timeline** κ projects both annotation sets onto a 1-second binary timeline (event / no event) and computes Cohen’s Kappa, capturing whether the two annotators agreed on *which moments* in the video are eventful. The **mean IoU** of greedily matched event pairs (threshold IoU ≥ 0.3) measures boundary precision: when both annotators flagged the same event, how similar were their timestamps? The **saliency** κ applies Cohen’s Kappa to the $-2 \dots +2$ ratings of matched pairs, revealing how much subjectivity influenced the importance scores.

Table 1: Inter-annotator agreement per video.

Video	#A	#B	timeline κ	saliency κ	mean IoU
33	10	9	0.148	0.600	0.655
34	8	8	0.478	0.357	0.703
35	9	10	0.384	0.226	0.554
36	15	11	0.254	−0.235	0.717

Table 1 tells a consistent story across three different lenses. The mean IoU of matched events is high everywhere (0.55–0.72), meaning that when both annotators flagged the same moment they tended to draw very similar time windows. The timeline κ is much lower (0.15–0.48), reflecting the fact that annotators disagreed less on *where* events are than on *how many* to mark. Video 33 is the starkest example: one annotator logged ten events by separating several short crowd reactions, while the other absorbed those into longer dance segments, producing a timeline κ of only 0.148. The saliency κ for video 36 (−0.235, below chance) is the most striking result in the table. The beekeeping video has no single narrative climax, so one annotator prioritised the beekeeper’s explanatory speech while the other rated visual demonstrations highest—two internally coherent but mutually incompatible mental models of what makes the video meaningful. Across all four videos, mean saliency κ is only 0.24, confirming that importance ratings are the loosest and most personal dimension of the annotation. For this reason we chose not to use saliency scores as a clip-selection signal

in Section 5: weighting by a metric the annotators disagreed on most would introduce more noise than signal.

For the rest of the pipeline we built a merged ground-truth set per video by greedily matching pairs of events with IoU ≥ 0.3 , intersecting their timestamps (taking the stricter, shorter window), and keeping the higher saliency score. Any event that did not match an event from the other annotator was kept as-is. This produced between 10 and 19 ground-truth events per video (15, 10, 13, and 19 respectively).

3 VLM EVENT DETECTION

We used **HuggingFaceTB/SmolVLM2-2.2B-Instruct** for automatic event detection. We chose it over Qwen2.5-VL-3B primarily because SmolVLM2’s processor accepts either a video path or a sequence of sampled frames natively, fits comfortably within our 8 GB GPU budget, and matches the demo notebook provided with the assignment, which means the grader can reproduce results without installing a Qwen licence.

Prompting strategies and coverage

We compared three prompting strategies. The first, **direct/events**, passes 6 uniformly sampled frames and asks the model to produce a `Salient event N: numbered list`. The second, **direct/timed**, uses the same frames and prompt but additionally requests a `[MM:SS-MM:SS]` timestamp for each event. The third, **caption-aggregate**, captions each frame individually and then feeds all captions back to SmolVLM2 with a merging prompt, following the strategy hinted at in the demo notebook. We measured coverage by encoding both the annotated and predicted event descriptions with sentence-transformers/all-MiniLM-L6-v2 and computing cosine similarity; an annotated event counts as covered if its best-matching predicted event has similarity ≥ 0.5 .

Table 2: VLM event-detection coverage. Strategy abbreviations: *DE* = direct/events, *DT* = direct/timed, *CA* = caption-aggregate. *cov@0.5*: fraction of annotated events covered at similarity ≥ 0.5 . *sim*: mean best-match similarity.

Video	Strat.	n_{ann}	n_{pred}	cov@0.5	sim
33	DE	15	4	0.00	0.235
33	DT	15	1	0.00	0.167
33	CA	15	2	0.13	0.392
34	DE	10	4	0.30	0.428
34	DT	10	5	0.10	0.361
34	CA	10	2	0.10	0.344
35	DE	13	4	0.08	0.325
35	DT	13	1	0.00	0.177
35	CA	13	1	0.00	0.114
36	DE	19	5	0.21	0.430
36	DT	19	1	0.00	0.018
36	CA	19	1	0.42	0.491

The clearest pattern in Table 2 is that *direct/timed* almost always fails. SmolVLM2 either refuses to produce timestamps entirely

or collapses the whole video into one spurious window such as `00:00-00:05`. Several prompt variants—“use MM:SS-MM:SS format”, “include explicit timestamps”, “each event must specify a start and end time”—produced the same failure, suggesting the model simply has no time-grounded visual supervision and cannot be prompted into acquiring it. By contrast, *direct/events* reliably extracts 4–5 events per video and achieves the best mean similarity on video 34 (0.428). The *caption-aggregate* strategy is interesting: it requires 7+ model calls per video and tends to collapse everything into one coarse description, but that one description can be a good semantic match when the video has a clear visual theme—hence 0.42 coverage on the beekeeping video, where every frame shares the same setting and subject. Overall, no strategy crosses the 0.5 coverage threshold on average, which is the expected zero-shot ceiling for a 2.2B model viewing only 6 frames. For practical use, *direct/events* is the right choice: it produces the most events, the highest mean similarity in three of the four videos, and is single-pass rather than requiring one model call per frame.

Failure modes and prompt history

Four failure patterns emerged consistently. Dialogue-driven events were missed entirely: because SmolVLM2 does not transcribe speech, an event annotated as “the speaker concludes by urging the students to let go of their fears” (video 33, 03:30–03:40, saliency +1) was detected only as “a bald man speaking into a microphone”—a mean embedding similarity of 0.18, well below the 0.5 threshold. Repetitive choreography (videos 33 and 34) collapses into a single event description even when the annotations distinguish 3–5 distinct formation changes, because the frames look nearly identical to the model. Domain-specific vocabulary in video 36 is partially lost: the model describes “a person in a white suit holding a frame” but never mentions “honey super” or “queen-bee cage”, which are the terms our annotations use. Finally, greedy decoding with many similar frames caused the model to repeat the same event description in a loop; we mitigated this by setting `repetition_penalty=1.3` and `no_repeat_ngram_size=4`, which eliminated obvious loops at the cost of occasionally shorter outputs.

On the prompt side, we found that generic instructions such as “describe the video” produced free-form paragraphs that our regex parser could not reliably split into discrete events. Switching to a templated `Salient event N: instruction` with a “Return only the numbered list” suffix raised the fraction of successfully parsed outputs from roughly 40 % to roughly 95 %. One non-obvious discovery was that including angle brackets (`< . . . >`) anywhere in the prompt caused the SmolVLM2 chat-template to emit zero tokens—an interaction with the model’s special-token vocabulary that took several iterations to diagnose.

4 MOMENT RETRIEVAL

We originally planned to use Lighthouse CG-DETR as our primary moment retrieval model, but the pre-trained checkpoint is absent from HuggingFace and the official download script on the current main branch returns HTTP 404. Rather than submit a non-functional integration, we report **two zero-shot CLIP variants**

that share the same retrieval code path but differ in backbone capacity: CLIP-B/32 (openai/clip-vit-base-patch32, 88 M parameters) and CLIP-L/14 (openai/clip-vit-large-patch14, 304 M parameters). For each query, frames are sampled at 1 Hz and encoded with the CLIP image tower; the query text is encoded with the text tower; per-frame cosine similarities are averaged over sliding windows of length $\{4, 8, 16, 32\}$ s; and the highest-mean window is returned after temporal NMS at IoU threshold 0.3. Both models additionally support paraphrase fusion, in which three deterministic rephrasings of each query (the original, “a video clip showing X”, and “footage of X”) are retrieved separately and their windows merged with NMS.

Spec-aligned evaluation: VLM events as queries

Following the assignment’s intended pipeline, we first ran CLIP-L/14 using the VLM *direct_events* descriptions as queries—17 events in total across the four videos—and measured IoU against the nearest ground-truth annotation window.

Table 3: CLIP-L/14 moment retrieval with VLM-detected events as queries, evaluated against the nearest GT annotation.

Video	VLM queries	mIoU	R@0.3
33	4	0.040	0.000
34	4	0.186	0.250
35	4	0.333	0.750
36	5	0.255	0.400
Overall	17	0.206	0.353

The overall mIoU of 0.206 is notably higher than the 0.113 reported below for annotated queries, but the comparison is not straightforward. The VLM produces only 4–5 coarse queries per video—descriptions like “crowd dancing” or “people inside a train”—whereas the annotated ground truth contains 10–19 fine-grained event descriptions per video. CLIP’s joint image-text space was trained on similarly coarse web captions, so the VLM queries land closer to the model’s natural operating regime. Video 35 illustrates both the strength and the limit of this approach: R@0.3 of 0.75 means three of the four VLM queries retrieved a reasonable window, but the missing query corresponded to the advertisement text reveal at the end of the video, which requires reading on-screen text rather than recognising visual content.

Fine-grained evaluation: annotated events as queries

To stress-test the retrieval with precise, human-authored descriptions, we ran all four method variants on the full 57 merged ground-truth events.

CLIP-L/14 is the clear winner, gaining roughly 27 % in mIoU and 50 % in R@0.3 over the smaller backbone at approximately three times the inference cost—still under 20 seconds per video on our hardware. Paraphrase fusion, however, consistently hurts performance: both backbones lose mIoU and R@0.3 when fusion is applied, though R@0.5 increases slightly for CLIP-L/14 (0.105 vs. 0.088). We

Table 4: Moment retrieval on all 57 annotated ground-truth events.

Method	mIoU	R@0.3	R@0.5
CLIP-B/32	0.089	0.140	0.070
CLIP-B/32 + fusion	0.082	0.105	0.053
CLIP-L/14	0.113	0.211	0.088
CLIP-L/14 + fusion	0.107	0.193	0.105

interpret this as fusion narrowing the predicted windows—which occasionally helps at the strict 0.5 threshold—while blurring the most discriminative text embedding, which hurts coarse alignment. A cross-attention architecture such as CG-DETR, which can process multiple query phrasings jointly, would likely profit from the same paraphrase inputs.

The per-category breakdown reveals where the model succeeds and where it breaks down fundamentally. Performance events—dancing, playing instruments—account for 29 of the 57 ground-truth events and achieve a mean IoU of 0.174, the highest of any category, because CLIP’s training data contains abundant examples of musicians and choreography. Beekeeping events (14 of 57) score 0.070: CLIP knows what bees and hives look like, but cannot distinguish “the beekeeper shows a frame” from “the beekeeper opens a hive”. Speech and dialogue events (9 of 57) score exactly 0.000 across all four methods. This is not a failure of any particular design choice—it is a fundamental limitation: CLIP has no access to audio or text within the video frame, so descriptions that depend on what someone is saying cannot be matched by visual similarity alone. The relationship between saliency and IoU is non-monotonic and revealing. The most important events (saliency +2, 21 events) achieve mean IoU 0.151; events rated +1 (14) drop to 0.091; events rated 0 (11) drop further to 0.027; then −1 events (8) recover to 0.127. The dip at zero and +1 captures exactly the transitional moments our annotations describe as “contextually necessary but not visually distinctive”—a conductor stepping forward, a crowd beginning to record on their phones. Those events are hardest to retrieve because their visual signature is generic.

5 VIDEO SUMMARY GENERATION

For each video we selected the predicted windows from CLIP-L/14, which achieved the best overall mIoU. We then applied a short post-processing pass: dropping any segment shorter than 1.5 s, merging segments separated by a gap of 1 s or less, sorting the result chronologically, and concatenating the clips into a single MP4 using the OpenCV writer (with an ffmpeg fallback where available).

Table 5: Generated summary statistics.

Video	Clips	Summary	Source
33	9	31 s	≈7:30
34	9	28 s	≈2:25
35	8	28 s	≈2:25
36	11	40 s	≈4:00

The summaries are concise—7 to 20 % of the source duration—because we clip each segment to its predicted window without adding any padding. Relevance is strong for visually distinctive content: videos 33–35 produce summaries dominated by dancing and instrument-playing segments, which are exactly the moments CLIP retrieves most accurately. The beekeeping summary is weaker on relevance because CLIP’s generic picks (hands holding objects, white protective suit) correspond to visually similar frames throughout the video rather than the specific named demonstrations the beekeeper is giving. Coverage suffers predictably from the speech problem: counting only events with $\text{IoU} \geq 0.3$, the summaries capture 21 % of all ground-truth events, and none of the nine speech-only events appear in any summary. Coherence is maintained locally by the chronological ordering and the gap-merging step, but global coherence in video 36 breaks down because the beekeeper’s explanations build on one another cumulatively, and extracting isolated visual clips loses the thread of his argument. The subjectivity disagreement has a direct trace here: because the two annotators assigned opposite saliency rankings to the same events in video 36, the merged ground truth treats narrative speech events and visual demonstrations as equally important. CLIP retrieves the visually distinctive ones and discards the rest, producing a summary that satisfies neither annotator’s conception of what the video is fundamentally about.

6 CRITICAL REFLECTIONS AND POSSIBLE IMPROVEMENTS

The result that stands out most from this assignment is not the IoU number itself but what drives it: the annotation process introduces more variance than any modelling decision we made. Our two annotators agreed well on boundaries (mIoU 0.65 on average) but substantially less on event count and saliency. Every IoU figure reported in Section 4 is measured against a merged ground truth that absorbs that disagreement, so the numbers should be read with that uncertainty in mind.

The single most impactful technical improvement would be adding automatic speech recognition. All nine speech-and-dialogue events in our ground truth achieve zero IoU under every method we tested, not because the retrieval architecture is wrong but because CLIP has no way to process spoken content. A lightweight ASR model such as Whisper-tiny (80 MB) would allow moment-retrieval queries to be matched against a transcript, and we expect this one change would have a larger effect on summary quality than any visual backbone upgrade.

Two improvements that operate within the existing visual pipeline are also worth pursuing. Paraphrase fusion did not help in our experiments, but we used only three static rephrasings. A model that generates semantically diverse paraphrases—rather than fixed string templates—might provide a genuine signal boost, especially for beekeeping events where the annotated descriptions use specialist vocabulary that CLIP’s training data may not associate strongly with the corresponding images. The saliency score could also be reintroduced at summary-generation time, not as an annotation metric but as a weighting signal produced by a small language model given the event description and the retrieved clip. Selecting

clips by predicted importance rather than treating all retrieved windows equally would give the summary a more coherent narrative arc, particularly for video 36 where the cumulative structure of the beekeeper’s explanation matters most.

Finally, the CG-DETR comparison we planned would have been instructive. Published results on QVHighlights report mIoU around 0.40 for CG-DETR, compared to 0.11 for our best CLIP variant. We implemented the integration code in `pipeline/moment_retrieval.py`, but the official pre-trained checkpoint is no longer publicly available, so we cannot report actual numbers. Once a working checkpoint is released, the comparison would be a direct test of whether the 25–30 IoU point gap holds on our informal, annotator-produced ground truth.

7 CONCLUSION

We built a complete event-detection-to-summary-video pipeline using zero-shot pretrained models, evaluated it against our own annotations with three inter-annotator agreement metrics and IoU across two moment-retrieval architectures, and ran failure analysis broken down by event category and saliency level. The dominant finding is that subjectivity in annotation is the largest single source of variance in the whole pipeline: the two annotators agreed on boundaries but disagreed on what counts as an event and how important each one is, and those disagreements propagate into every downstream metric. The second dominant finding is that zero-shot visual retrieval fails completely on speech events, and the practical path forward for this kind of pipeline runs through automatic speech recognition rather than through larger or better-tuned visual encoders.